JBCN STARS

Scholastic Tournament for Advanced Mathematics and Reasoning Skills
Practice Paper (unofficial, modelled on the 2023–2025 papers)

TEAM ROUND

Level: Junior **Pattern: Problem solving**

Maximum marks: 100 **Duration 60 minutes**

Instructions:

- This document consists of 10 questions in increasing order of difficulty.
- The team has 60 minutes to discuss and solve them.
- Each question carries 10 marks. There is no penalty for a wrong answer.
- Keep your voices soft while discussing.
- Use blank sheets for working of any kind.
- Calculators are not permitted.
- Give answers in the exact form or as specified (include units).
- At the 55-minute reminder, the team captain uses the last 5 minutes to write the answers on the response sheet.

Requirements:

Response sheet, pen, pencil, eraser and scratch paper

On competition day the Junior level does the Team Round first (08:00–09:15), then the Individual Round (09:35–10:45) after the snack break. Practise in the same order.

Team Round – Junior Level

1. Riya sold 40 bookmarks and Kabir sold 60 bookmarks. All the bookmarks were sold at the same price. Together they received ₹750. How much money did Kabir receive?

2. The volume of a cube is 512 cm^3. What is the total surface area of the cube?

3. In a football league a team gets 3 points for a win, 1 point for a draw and 0 points for a loss. At the end of the season, Pi United had 52 points after playing 30 games. They drew 7 games. How many games did they lose?

4. The sum of the squares of two consecutive positive whole numbers is 313. What is the sum of the two numbers?

5. A bus is carrying 54 people, all men or women. When 6 men get off, the ratio of men to women on the bus becomes 1 : 3. How many women are on the bus?

6. A snail is at the bottom of a pole 20 m tall. Every day it climbs up 4 m, and every night it slips down 3 m. On which day does the snail first reach the top of the pole?

7. Pipes A and B can fill a tank in 4 hours and 6 hours respectively. Pipe C can empty the full tank in 8 hours. If all three pipes are opened together on an empty tank, how long will it take to fill the tank? Give your answer in hours and minutes, rounded to the nearest minute.

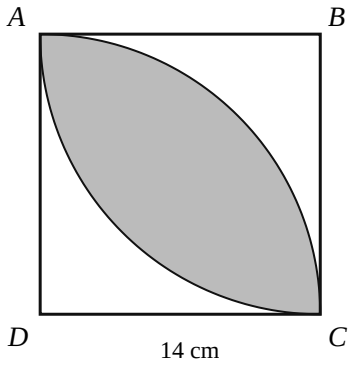
8. The average age of a group of children is 12.5 years. A teacher aged 38 years joins the group, and the average age of the group rises to 14 years. How many children were in the group?

9. a and b are positive whole numbers with $a < b$ and

$$a + ab + b = 34$$

Find $a + b$.

10. *ABCD* is a square of side 14 cm. Two quarter circles of radius 14 cm are drawn inside the square, one with centre *B* and one with centre *D*. Find the area of the shaded leaf-shaped region where they overlap. (Take $\pi = \frac{22}{7}$.)



Team Round – Response Sheet

Team ID: _____

Q	Answer (with units)
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Score: _____ / 100

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INDIVIDUAL ROUND

Level: Junior **Pattern: MCQ**

Maximum marks: 120 **Duration 60 minutes**

Instructions:

- This round consists of 30 multiple choice questions.
- Mark your answers on the response sheet.
- Select only one answer for each question.
- Every correct answer is awarded 4 points.
- Every incorrect answer gets −1 point. A blank answer gets 0.
- You are not required to show working.
- Calculators are not allowed.
- Time allotted for this round is 60 minutes (1 hour).

Requirements:

Response sheet, pen, pencil, eraser and scratch paper

Tip: with −1 for a wrong answer, guess only when you can rule out at least two options.

Individual Round – Junior Level

1. For any numbers a and b, $a \bigstar b = 2a + ab$. What is $(2 \bigstar 3) \bigstar 1$?

A. 24 **B.** 26 **C.** 28 **D.** 30 **E.** 36

2. How many two-digit numbers have digits that add up to 9?

A. 8 **B.** 9 **C.** 10 **D.** 11 **E.** 12

3. A movie starts at 4:45 pm and lasts 2 hours 35 minutes. At what time does it end?

A. 7:10 pm **B.** 7:15 pm **C.** 7:20 pm **D.** 7:25 pm **E.** 7:30 pm

4. 15% of 300 is equal to 30% of

A. 45 **B.** 90 **C.** 100 **D.** 120 **E.** 150

5. How many of these five numbers are perfect squares?

$$2^4 \quad 3^5 \quad 4^3 \quad 5^6 \quad 6^5$$

A. 1 **B.** 2 **C.** 3 **D.** 4 **E.** 5

6. The sum of four consecutive odd numbers is 72. What is the largest of these numbers?

A. 19 **B.** 21 **C.** 23 **D.** 25 **E.** 17

7. Which of these fractions lies between $\frac{3}{5}$ and $\frac{2}{3}$?

A. $\frac{5}{8}$ **B.** $\frac{7}{12}$ **C.** $\frac{11}{16}$ **D.** $\frac{4}{7}$ **E.** $\frac{7}{10}$

8. Peter the penguin went fishing. He realised that if he had caught four times as many fish as he actually did, he would have had 27 more fish. How many fish did Peter catch?

A. 6 **B.** 7 **C.** 8 **D.** 9 **E.** 12

9. A sheet of paper is folded in half, then folded in half again, and so on, until it has been folded in half 6 times. How many layers of paper are there in the folded sheet?

A. 12 **B.** 24 **C.** 32 **D.** 36 **E.** 64

10. In a class of 40 children, the ratio of boys to girls is 3 : 5. How many girls must leave the class so that the numbers of boys and girls are equal?

A. 5 **B.** 8 **C.** 10 **D.** 12 **E.** 15

11. The mean of five numbers is 20. Tara adds 2 to the first number, 4 to the second, 6 to the third, 8 to the fourth and 10 to the fifth. What is the mean of the new numbers?

A. 26 **B.** 30 **C.** 50 **D.** 21 **E.** 24

12. One angle of a triangle is 40°. The other two angles are in the ratio 2 : 5. What is the largest angle of the triangle?

A. 60° **B.** 80° **C.** 90° **D.** 95° **E.** 100°

13. What is the units digit of 7^{2025}?

A. 7 **B.** 9 **C.** 3 **D.** 1 **E.** 5

14. The length of a rectangle is increased by 20% and its width is decreased by 20%. The new area is what percent of the old area?

A. 100% **B.** 96% **C.** 104% **D.** 90% **E.** 80%

15. A square has a perimeter of 36 cm. A rectangle has the same area as the square and a width of 3 cm. What is the perimeter of the rectangle?

A. 48 cm **B.** 54 cm **C.** 56 cm **D.** 60 cm **E.** 66 cm

16. In a certain month, three of the Saturdays fall on even-numbered dates. On what day of the week is the 25th of that month?

A. Sunday **B.** Monday **C.** Tuesday **D.** Thursday **E.** Friday

17. How many positive whole numbers are factors of 2025?

A. 9 **B.** 12 **C.** 15 **D.** 16 **E.** 18

18. Asha, Bina and Chetan all have different heights. Exactly **one** of these statements is true:

I. Asha is the tallest.
II. Bina is not the tallest.
III. Chetan is not the shortest.

Arrange them from tallest to shortest.

A. Bina, Chetan, Asha
B. Bina, Asha, Chetan
C. Chetan, Bina, Asha
D. Asha, Chetan, Bina
E. Chetan, Asha, Bina

19. 3 apples weigh the same as 2 oranges, and 4 oranges weigh the same as 3 melons. How many apples weigh the same as 6 melons?

A. 8 **B.** 9 **C.** 10 **D.** 12 **E.** 16

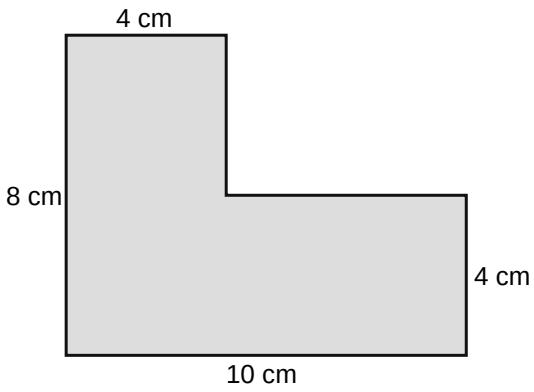
20. A tap drips 1 mL of water every 3 seconds. How many hours will it take to fill a 2.4 litre bottle?

A. 1 **B.** 2 **C.** 3 **D.** 4 **E.** 6

21. A car travels at a constant speed of 40 km/h. A motorbike starts 30 km behind the car and catches up with it in 1½ hours. What is the speed of the motorbike?

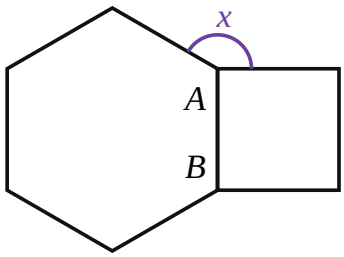
A. 50 km/h **B.** 55 km/h **C.** 60 km/h **D.** 65 km/h **E.** 70 km/h

22. All the corners of the shape below are right angles. What is its perimeter?



A. 28 cm **B.** 30 cm **C.** 32 cm **D.** 34 cm **E.** 36 cm

23. A regular hexagon and a square share the side AB, as shown. What is the size of angle x?



A. 120° **B.** 135° **C.** 150° **D.** 160° **E.** 210°

24. How many different four-digit numbers can be made by rearranging the digits of 2205? (A number cannot start with 0.)

A. 6 **B.** 7 **C.** 8 **D.** 9 **E.** 12

25. A 3 × 3 × 3 cube is painted red on all its outside faces and then cut into 27 small cubes. How many small cubes have exactly two red faces?

A. 1 **B.** 6 **C.** 8 **D.** 10 **E.** 12

26. Two opposite corners of a cube are cut off, each cut creating a new triangular face. How many edges does the new solid have?

A. 18 **B.** 15 **C.** 20 **D.** 21 **E.** 24

27. Six different points are marked on a circle. How many different triangles can be drawn whose three corners are all among these points?

A. 12 **B.** 15 **C.** 18 **D.** 20 **E.** 24

28. In a magic square, every row, column and diagonal has the same sum. The magic square below uses each of the whole numbers from −4 to +4 exactly once. What is the value of Y?

-3		Y
$+4$		
-1		$+3$

A. −2 **B.** −1 **C.** 0 **D.** +2 **E.** +1

29. Asha, Bina, Chetan, Dev and Esha sit in a row of five seats. Asha sits in the leftmost seat. Chetan sits next to both Dev and Esha. In how many different orders can the five friends sit?

A. 2 **B.** 4 **C.** 6 **D.** 8 **E.** 12

30. How many whole numbers from 1 to 100 are divisible by 3 or by 5, but **not** by both?

A. 41 **B.** 47 **C.** 53 **D.** 35 **E.** 39

Individual Round – Response Sheet

Name / Participation ID: _____ Circle one letter per question.

Q	Answer	Q	Answer	Q	Answer
1	A B C D E	11	A B C D E	21	A B C D E
2	A B C D E	12	A B C D E	22	A B C D E
3	A B C D E	13	A B C D E	23	A B C D E
4	A B C D E	14	A B C D E	24	A B C D E
5	A B C D E	15	A B C D E	25	A B C D E
6	A B C D E	16	A B C D E	26	A B C D E
7	A B C D E	17	A B C D E	27	A B C D E
8	A B C D E	18	A B C D E	28	A B C D E
9	A B C D E	19	A B C D E	29	A B C D E
10	A B C D E	20	A B C D E	30	A B C D E

Correct ____ × 4 − Wrong ____ × 1 = Score _____ / 120